

Text, Screenshots, or Both?

Representation Value and Task Routing in Web Agents

Anonymous ACL submission

Abstract

A web agent can read a page as text, look at a screenshot, or do both, and deployments usually fix one choice for every task. We measure that choice across three sites of VisualWebArena and WebArena with four vision-language backbones, using text-only variants as controls and rerunning identical conditions to see which differences survive. Three findings separate questions that are usually merged. First, no observation choice is best everywhere, and the screenshot has a real, reproducible value that is predictable in advance on one site: a zero-cost rule over the task text picks out classifieds tasks on which adding the screenshot alone raises success by 25–30 points for the two strongest backbones (11 for a 4B model), and the gap reappears on a full rerun. Second, which representation suits which task is a genuine property of the task, beyond its difficulty, yet a single run measures it with reliability of only 0.31–0.50. Third, routers that choose a representation per task, with their operating point fixed on training data, do not convert this signal into a gain over the best fixed choices and their random mixtures. We describe the rerun-controlled protocol that keeps these three quantities apart.

1 Introduction

An agent that operates a website has to perceive it, and the two standard ways are complementary in principle: the page’s text (an accessibility tree) is precise and cheap, a screenshot shows what the text omits. A third option fuses them, overlaying numbered marks on the screenshot and listing the marked elements as text (Yang et al., 2023; Koh et al., 2024). Practitioners typically pick one of the three for a whole deployment. The question this paper asks is the one that choice raises: *is the choice worth making per task, and can it be made?*

Answering it requires keeping apart three quantities that a single leaderboard number merges:

1. **Representation value:** does the screenshot change success, holding everything else fixed?
2. **Preference reliability:** when one representation beats another on a task, is that a stable property of the task, or a property of one run?
3. **Routing gain:** can a policy that chooses per task, using only what is known before it runs, beat the fixed choices?

A large hindsight ceiling — pick, for each task, whichever representation happened to succeed — answers none of them, because it rewards run-to-run luck as much as structure. We therefore rerun identical conditions and read every comparison against what a rerun alone changes.

Findings. (i) No choice wins everywhere: the fused mode leads on classifieds, text leads on WebArena reddit, and the order reverses across backbones (§3). (ii) The screenshot has a specific value that can be isolated and predicted: with the text payload and prompt held fixed, adding the screenshot raises success on the classifieds tasks a zero-cost intent rule flags by 25–30 points for the two strongest backbones (11 for a 4B model), against 5–10 on the other tasks; the effect reproduces on a full rerun and is largest for the strongest backbone. On reddit the same rule finds nothing (§4). (iii) Per-task preferences between representations exist beyond task difficulty — a no-interaction null conditioned on difficulty is rejected in every fully rerun cell — but a single run measures them with reliability 0.31–0.50; two to three runs per task are needed to pass 0.5 (§5). (iv) Routers that fix their operating point on training data do not beat the fixed representations and their random mixtures; the room that exists above that frontier comes from task difficulty, not from representation fit, and is not captured without looking at test outcomes (§6).

Contributions. A rerun-controlled measurement of text, screenshot and fused observation across

three sites and four backbones, with text-only ablations that isolate the screenshot; a matched estimate of the screenshot’s value and an ex-ante rule that predicts it on one site; a test of whether per-task representation preferences are real, with a decision study for how many runs make them learnable; and a deployable-policy comparison against fixed choices *and their random mixtures*, which is the comparison a routing claim has to win (§7).

2 Setup

Sites and tasks. We use three sites of VisualWebArena (Koh et al., 2024) — classifieds (224 scored tasks), reddit (203) and shopping (432) — and the reddit site of WebArena (Zhou et al., 2024) (104 tasks, those all observation modes ran). VWA-reddit and WA-reddit are different task sets on the same application, so they are not two independent benchmarks. Scored sets are the collected tasks minus protocol exclusions written as rules over task configuration (Appendix A).

Backbones. B0 = Qwen3-VL-235B (API-served), B1 = Qwen3-VL-4B and B2 = Gemma-3-4B (both served locally), and B5 = GPT-5.6 (API, classifieds only). A *cell* is one site × backbone pair; the eight cells with every observation mode are VWA-classifieds and VWA-reddit × {B0, B1, B2} and WA-reddit × {B0, B1}. B5-classifieds and the two shopping cells extend them (§4, §6) and are never pooled with the eight.

One plain agent. One model call per step emits a thought, an action and its arguments; the prompt holds the instruction, the mode’s system prompt, the last eight steps and the current observation. There is no planner, memory, reflection or retry policy, so a difference between conditions is a difference of observation, not of scaffold. Episodes stop at 30 steps.

Observation modes. The deployment question has three answers: **DOM** (the page’s accessibility tree as text), **Vision** (the raw screenshot, with an empty text payload) and **SoM** (a set-of-marks screenshot plus the list of marked elements as text; Yang et al., 2023). Three further modes carry no screenshot and serve as controls; the one used throughout is **P-SoM**, which is SoM with the screenshot removed — same marks, same prompt — so SoM–P-SoM isolates the image (Appendix B lists all six). “Text” modes withhold the *page*

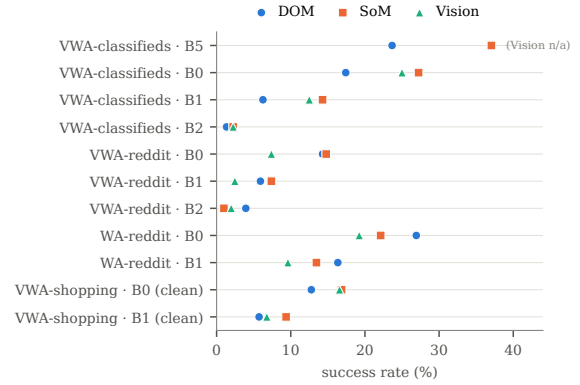


Figure 1: Success rate of the three deployment choices per cell. Shopping cells drop the tasks touched by a known harness defect in any mode (§2, Appendix D); B5’s Vision run is excluded for a coordinate-contract defect. Source: routing_three_arm, routing_extension_cells.

screenshot only: a reference image that is part of the task (“find this item”) is given in every mode.

Reruns. We reran identical conditions, 24 registered same-condition pairs in all; in three cells — VWA-classifieds B0, VWA-reddit B0 and WA-reddit B1 — every mode was run twice. A rerun uses the same tasks, code and site reset, so what it changes is what one run of the agent does not determine.

Scoring and cost. The evaluators are binary. Cost is the per-episode billed cost: an API invoice for B0 and B5, a token-priced electricity estimate for B1 and B2. The two kinds are ordered within a cell, never pooled across; §6 repeats the routing analysis under GPU-time and wall-clock cost.

3 No observation choice is best everywhere

Figure 1 shows the three deployment choices side by side. On classifieds the fused mode is the best of the three for B0, B1 and GPT-5.6 (B2 solves too few tasks to order the modes), and the margin over text is not small: SoM–DOM is +9.82 pp [+3.57, +16.07] for B0 and +8.04 pp [+3.57, +12.95] for B1 (paired task bootstrap). On WebArena reddit the order flips and text leads (SoM–DOM −4.81 pp for B0 and −2.88 pp for B1, both intervals crossing zero), and on VWA-reddit the weakest backbone does significantly worse with the fused mode (−2.96 pp [−5.91, −0.49]). On shopping the fused mode is best or tied with Vision (Figure 1).

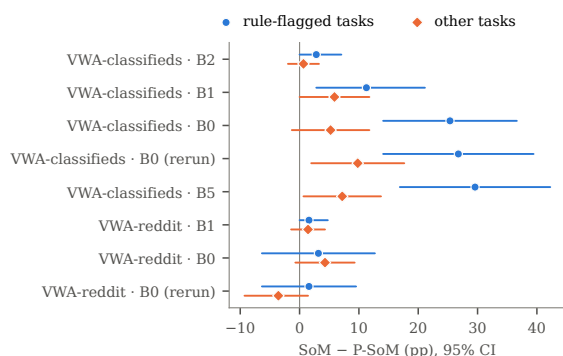


Figure 2: Value of the screenshot alone (SoM–P-SoM: same marks, same prompt) on the tasks a zero-cost intent rule flags, against the other tasks. Classifieds is out-of-sample for the rule. “Rerun” is a full repeat of both modes. VWA-reddit B2 is omitted: no mode solves a flagged task. Source: visual_intent_routing.

Two cautions keep this descriptive. Cells on the same site share their tasks, so the eight are not eight independent replications. And the screenshot-free modes are four, not one: a maximum over four text variants against a single screenshot mode would be a selection effect, so the comparison here takes one representative per deployment class, as practitioners do.

What this section establishes is only that the choice matters and depends on the workload. Whether it can be made per task is the subject of the rest of the paper.

4 The screenshot has a value that can be isolated and predicted

A matched contrast. Comparing Vision with DOM changes two things at once — the image and the text the agent receives. SoM and P-SoM differ only in the annotated screenshot, so their difference is the value of the image alone.

An ex-ante rule. A regular expression over the task intent flags tasks that ask about an image, a photo, a colour or a count of things shown, excluding tasks that already carry a reference image (Appendix C). It reads only the task configuration: no model call and no episode. It was written for reddit, so its classifieds hits were never inspected and classifieds is out-of-sample.

Result. On the 71 classifieds tasks it flags, the screenshot raises success by +25.35 pp [+14.08, +36.62] for B0, against +5.23 pp on the other 153 tasks (Figure 2). The effect reproduces on a full

rerun of both modes (+26.76 pp [+14.08, +39.44], from 23 against 4 successes), and it grows with capability: +2.82 pp for B2, +11.27 pp for B1, +25.35 pp for B0 and +29.58 pp [+16.90, +42.25] for GPT-5.6. On VWA-reddit the same rule flags 63 tasks and separates nothing: +3.17 pp on flagged tasks against +4.29 pp elsewhere for B0, +1.59 pp against −3.57 pp on its rerun. On shopping the flagged tasks carry at most five successes per mode, too few to read, and on WebArena the rule flags only 5 of 104 tasks.

Reading. Where the screenshot pays off on classifieds is predictable before anything runs, and it pays off more for stronger models; on reddit it is not. The screenshot’s value is therefore real and partly foreseeable, but site-specific: it is a statement about a kind of task, not a general law. It is also not a router. On these tasks the fused mode already contains the screenshot, so switching between DOM and Vision by this rule cannot beat simply using SoM; what remains open is whether *individual* tasks have stable preferences beyond such task types, which is the question of §5.

5 Per-task preferences are real, but one run cannot measure them

[TODO: chunk 2: no-interaction null (three arms primary, six in appendix), D-study, same-task persistence across reruns.]

6 Deployable routers do not convert the signal

[TODO: chunk 2: frontier with random mixtures, deployable cross-fitted policy and bounds, template/rerun, cost bases, B5 and shopping.]

7 A protocol for multi-representation claims

[TODO: chunk 2: the four-step protocol.]

8 Related work

[TODO: chunk 3: only citations already in paper.bib.]

Limitations

[TODO: chunk 3: required by ARR.]

References

Jing Yu Koh, Robert Lo, Lawrence Jang, Vikram Duvvur, Ming Chong Lim, Po-Yu Huang, Graham Neubig, Shuyan Zhou, Ruslan Salakhutdinov, and Daniel Fried. 2024. [Visualwebarena: Evaluating multimodal agents on realistic visual web tasks](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.

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A Task universes and exclusions

[TODO: chunk 3]

B The six observation modes

[TODO: chunk 3]

C The intent rule

[TODO: chunk 3]

D Harness defects and how cells were filtered

[TODO: chunk 3]